

ASSIGNMENT 1 – MINI PROJECT

Console Record-Management Application

LIBRO – LIBRARY MANAGEMENT SYSTEM

MCA Semester I – Python Programming

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1. Introduction

LIBRO – Library Management System is a menu-driven, console-based application developed in Python for Assignment 1 of MCA Semester I – Python Programming. It manages library books and borrowing transactions through a simple terminal interface. The application supports adding, viewing, searching, updating and deleting books, along with issuing, returning and tracking transactions.

CSV files provide persistent storage, so data remains available after the program is closed. The project demonstrates the core Python concepts specified in the assignment without using a database or graphical interface.

2. Objectives

- Develop a fully working console-based record-management application.
- Provide a menu-driven interface for library operations.
- Add, view, search, update and delete book records.
- Issue and return books while maintaining transaction history.
- Use appropriate Python data types and data structures.
- Organize logic using functions.
- Handle invalid input and file-related errors.
- Store and retrieve records using CSV File I/O.

3. Problem Statement

A small library needs a simple system to maintain book records and track books issued to students. The system should provide common record-management operations through a console interface and preserve information between sessions. LIBRO addresses this requirement using Python and CSV-based storage.

4. Technologies Used

Technology / Tool	Purpose
Python	Application development
CSV	Persistent storage of books and transactions
Python csv module	Reading and writing CSV files
VS Code	Development, execution and testing
GitHub	Source-code and documentation submission

5. Python Concepts Used

Concept	Use in LIBRO
Data Types and Variables	Strings, integers, lists, dictionaries and variables store records.
Conditional Statements	if / elif / else handle choices, validation and decisions.
Loops	while loops handle menus and repeated input; for loops process records.
Functions	Separate functions handle login, file operations and library features.
Exception Handling	try / except handles invalid numeric input and file-related errors.
File I/O	CSV files are read and written for persistent storage.
Menu-driven Design	A numbered menu lets the user select operations.

6. Main Features

Feature	Description
Login	Authenticates the librarian.
Dashboard	Shows total books, available copies, currently issued copies and active borrowers.
Add Book	Adds a new book after validation.
View All Books	Displays book records in a formatted table.
Search Book	Searches by Book ID, Book Name, Author, Genre or Category.
Update Book	Changes existing book details or quantity.
Delete Book	Deletes a book after confirmation and checks whether it is currently issued.

Feature	Description
Issue Book	Issues an available book and records the transaction.
Return Book	Returns an issued book and updates availability.
Issue History	Displays issue and return transactions.
Logout / Exit	Ends the session or application.

7. System Working

1. Start the program
2. Initialize/check the CSV files
3. Login
4. Display the main menu
5. Select an operation
6. Read required data from CSV files
7. Perform the operation
8. Save changed data to CSV
9. Return to the main menu
10. Logout or Exit

8. Data Files

File	Fields
books.csv	Book ID, Book Name, Author, Genre, Category, Quantity
transactions.csv	Student ID, Student Name, Book ID, Book Name, Date, Status

Quantity represents currently available copies. Issuing a book decreases the quantity by one; returning it increases the quantity by one. transactions.csv stores the student, book, date and transaction status.

9. Function Reference

Function	Purpose
setup()	Creates CSV files with sample data if they do not exist.
read()	Reads records from a CSV file.
write()	Writes records to a CSV file.
books()	Reads book records and converts Quantity to an integer.
header()	Prints formatted section headings.
table()	Displays records in a formatted table.
text()	Validates non-empty text input.
number()	Validates non-negative integer input.
find()	Finds a book by Book ID.
login()	Checks librarian credentials.
dashboard()	Calculates and displays library summary.
add_book()	Adds a new book.
view_books()	Displays all books.
search_book()	Searches books using a selected field.
update_book()	Updates an existing book.
delete_book()	Deletes a book after validation and confirmation.
issue_book()	Issues a book and creates an Issued transaction.
return_book()	Returns a book and changes its transaction to Returned.
history()	Displays transaction history.
menu()	Controls the main menu.
main()	Starts setup, login and application flow.

10. Exception Handling

- ValueError is handled for invalid whole-number input.
- FileNotFoundError is handled when a CSV file is missing.
- Other file/runtime errors are caught and displayed.
- Empty text and negative quantities are rejected by validation.
- Invalid menu choices display an error instead of terminating the program.

11. Testing

Test Case	Expected Result	Result
Valid login	Main menu displayed	Pass
Dashboard	Library summary displayed	Pass
Add book	New book saved to books.csv	Pass
View books	Book records displayed	Pass
Search by author	Matching book displayed	Pass
Update book	Selected details updated	Pass
Issue book	Quantity decreases and transaction is recorded	Pass
Return book	Quantity increases and transaction becomes Returned	Pass
Issue history	Transactions displayed	Pass
Delete book	Book removed after confirmation	Pass
Restart application	Saved CSV data remains	Pass

12. Screenshots

Screenshots captured during actual testing of the working console application.

Figure 1: Login and Main Menu

```
C:\Users\sagar\OneDrive\Desktop\PythonMCA>python libro.py

=====
LIBRO - LIBRARY MANAGEMENT SYSTEM
=====
Please login to continue.

Username: admin
Password: admin123

Login successful!

=====
LIBRO - MAIN MENU
=====
1. Dashboard
2. Add Book
3. View All Books
4. Search Book
5. Update Book
6. Delete Book
7. Issue Book
8. Return Book
9. Issue History
10. Logout
11. Exit

Enter choice (1-11):
```

Figure 2: Dashboard

```
=====
LIBRO - MAIN MENU
=====
1. Dashboard
2. Add Book
3. View All Books
4. Search Book
5. Update Book
6. Delete Book
7. Issue Book
8. Return Book
9. Issue History
10. Logout
11. Exit

Enter choice (1-11): 1

=====
DASHBOARD
=====
Total Books      : 53
Available Copies : 48
Currently Issued : 5
Active Borrowers : 5

Press Enter to continue...
```

Figure 3: View All Books

```
LIBRO - MAIN MENU
1. Dashboard
2. Add Book
3. View All Books
4. Search Book
5. Update Book
6. Delete Book
7. Issue Book
8. Return Book
9. Issue History
10. Logout
11. Exit

Enter choice (1-11): 3

=====
ALL BOOKS
=====
Book ID | Book Name | Author | Genre | Category | Quantity | Status
-----|-----|-----|-----|-----|-----|-----
B001 | The Alchemist | Paulo Coelho | Fiction | Novel | 4 | Available
B002 | A Brief History of Time | Stephen Hawking | Science | Non-Fiction | 3 | Available
B003 | Introduction to Algorithms | Thomas Cormen | Computer Science | Textbook | 2 | Available
B004 | Wings of Fire | A.P.J. Abdul Kalam | Biography | Non-Fiction | 4 | Available
B005 | Harry Potter | J.K. Rowling | Fantasy | Novel | 1 | Available
B006 | Clean Code | Robert C. Martin | Computer Science | Textbook | 3 | Available
B007 | Atomic Habits | James Clear | self-help | Non-fiction | 4 | Available
B008 | Kingdom of Ashes | Liam Night | Fantasy | Novel | 2 | Available
B009 | The Metamorphosis | Franz Kafka | Fiction | Novel | 0 | Out of Stock
B010 | Pride and Prejudice | Jane Austen | Romance | Novel | 4 | Available
B011 | Whispers of the Forgotten | Mira Blake | Mystery | Novel | 5 | Available
B012 | Crime and Punishment | Fyodor Dostoevsky | Psychological Fiction | Novel | 3 | Available
B013 | The Hunger Games | Suzanne Collins | Dystopian | Fiction | 3 | Available
B014 | It ends with us | Colleen Hoover | Romance | Novel | 2 | Available
B015 | Operating System Concepts | Abraham Silberschatz | Operating System | Textbook | 8 | Available

Press Enter to continue...|
```

Figure 4: Search Book by Author

```
LIBRO - MAIN MENU
1. Dashboard
2. Add Book
3. View All Books
4. Search Book
5. Update Book
6. Delete Book
7. Issue Book
8. Return Book
9. Issue History
10. Logout
11. Exit

Enter choice (1-11): 4

=====
SEARCH BOOK
=====
1. Book ID 2. Book Name 3. Author 4. Genre 5. Category
Choose field (1-5): 3
Search: Franz Kafka
Book ID | Book Name | Author | Genre | Category | Quantity | Status
-----|-----|-----|-----|-----|-----|-----
B009 | The Metamorphosis | Franz Kafka | Fiction | Novel | 0 | Out of Stock

Press Enter to continue...|
```

Figure 5: Issue Book

```
=====
LIBRO - MAIN MENU
=====
1. Dashboard
2. Add Book
3. View All Books
4. Search Book
5. Update Book
6. Delete Book
7. Issue Book
8. Return Book
9. Issue History
10. Logout
11. Exit

Enter choice (1-11): 7

=====
ISSUE BOOK
=====
Book ID: B014
Student ID: S007
Student Name: Shawn Thomas
Book issued successfully.

Press Enter to continue...|
```

Figure 6: Return Book

```
Enter choice (1-11): 8

=====
RETURN BOOK
=====
Student ID: S007
Book ID | Book Name      | Date
-----|-----|-----
B007    | Atomic Habits    | 2026-09-19
B014    | It ends with us | 2026-09-20
Book ID to return: B014
Book returned successfully.

Press Enter to continue...|
```

Figure 7: Issue History

Enter choice (1-11): 9

=====

ISSUE HISTORY

=====

Student ID	Student Name	Book ID	Book Name	Date	Status
S001	Rahul Sharma	B005	Harry Potter	2026-09-01	Returned
S002	Priya Singh	B003	Introduction to Algorithms	2026-09-05	Issued
S010	Samiksha Pawar	B011	Whispers of the Forgotten	2026-09-19	Returned
S003	Ramesh Jha	B013	The Hunger Games	2026-09-19	Issued
S005	Deeya Mathur	B004	Wings of Fire	2026-09-19	Issued
S006	Farida Khan	B004	Wings of Fire	2026-09-19	Returned
S006	Farida Khan	B009	The Metamorphosis	2026-09-19	Issued
S007	Tanvi Sharma	B007	Atomic Habits	2026-09-19	Issued
S007	Shawn Thomas	B014	It ends with us	2026-09-20	Returned

Press Enter to continue...

13. Project Structure

LIBRO-Library-Management-System/

- libro.py
- books.csv
- transactions.csv
- README.md
- Assignment1_Report.pdf
- screenshots/
 - login.png
 - dashboard.png
 - view_books.png
 - search_book.png
 - issue_book.png
 - return_book.png
 - issue_history.png

14. How to Run

1. Install Python 3.x.
2. Open the project folder in VS Code or a terminal.
3. Open a terminal in the project folder.
4. Run: **python libro.py**
5. Enter the librarian credentials.
6. Use the numbered menu to perform operations.

Demo login: Username = admin | Password = admin123

15. Conclusion

The LIBRO Library Management System integrates variables and data types, conditional statements, loops, functions, exception handling and File I/O into a working console application. It provides book management, issue and return operations, transaction history, input validation and persistent CSV storage. The application was run and tested through its major features before submission.

16. GitHub Submission Details

Repository Name: LIBRO-Library-Management-System

GitHub Repository Link: To be added after repository creation

Student: Samiksha Pawar

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